# Numerical Behavior of the Bratu Problem Near Criticality

**Existence, Nonexistence, and Global Consistency Near a Saddle-Node Fold**

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## SIRPY · Problem 8 — A Case Study in Solution Sensitivity and Numerical Artifacts

**An ongoing series. We are building a public library of problems SIRPY has solved — each entry is one problem, the exact input we handed the solver, the exact output it returned, and an honest read of what happened. No slides. No hand-waving. Numbers you can check.**

## Introduction

The Bratu problem is a classical nonlinear boundary-value problem

$$y'' + \lambda e^{y} = 0, \qquad y(0) = 0, \qquad y(1) = 0.$$

It is a standard benchmark in nonlinear numerical analysis because it exhibits a saddle-node bifurcation.

A critical parameter exists near

$$\lambda_c \approx 3.513830719.$$

Theoretical bifurcation analysis predicts:

- $0 < \lambda < \lambda_c$: two solutions exist,
- $\lambda = \lambda_c$: the two branches merge,
- $\lambda > \lambda_c$: no classical solution exists.

This benchmark studies how SIRPY behaves as the fold is approached.

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## Methodology

The default SIRPY solve was

```
for lam in (3.5135, 3.5138, 3.513830719):

    r = solve_bvp(
        order=2,
        f_str=f"-{lam}*exp(y)",
        x0=0,
        x1=1,
        left_bc={0:0.0, 1:"gamma"},
        right_bc={0:0}
    )
```

The solver automatically:

- estimates series reliability,
- performs bracket-march shooting,
- verifies the ODE by substitution,
- measures boundary residuals,
- measures interface continuity,
- attempts nonexistence certification when appropriate.

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## Case 1: Subcritical Parameter

$\lambda = 3.5135$

### Solver Output

```
INFO: verification by substitution - VERIFIED

Equation residual:
6.38e-10
```

```
Relative residual:
6.19e-11

Boundary residual:
4.44e-16

Interface continuity:
1.6e-11

Segments:
6
```

### Interpretation

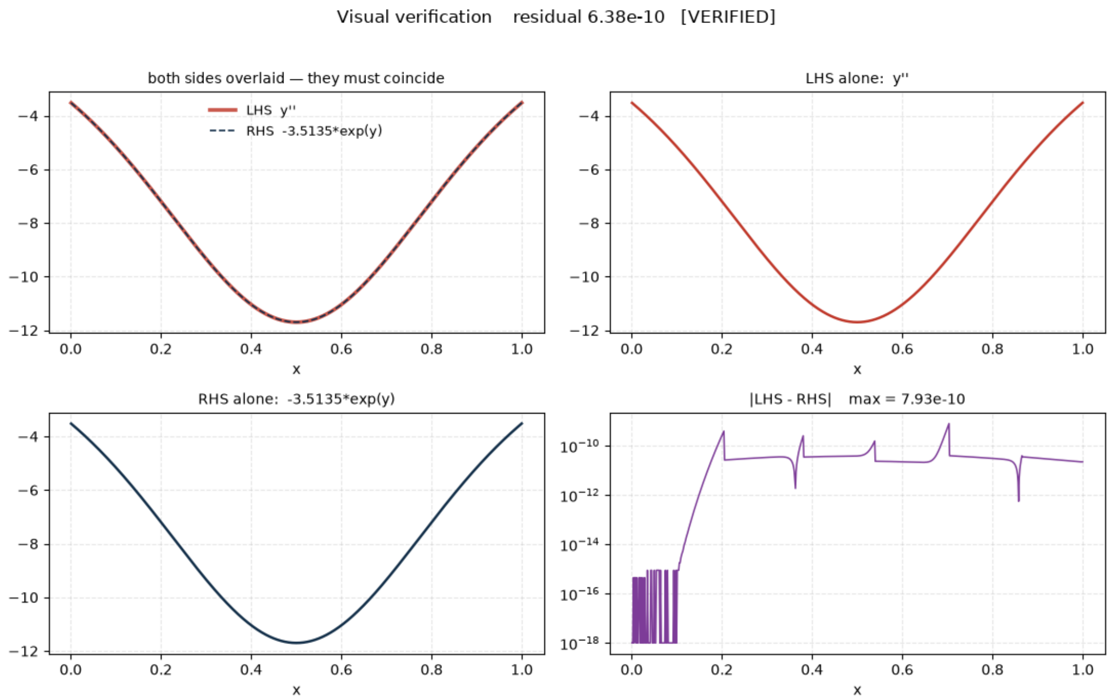
This is a straightforward successful solve.

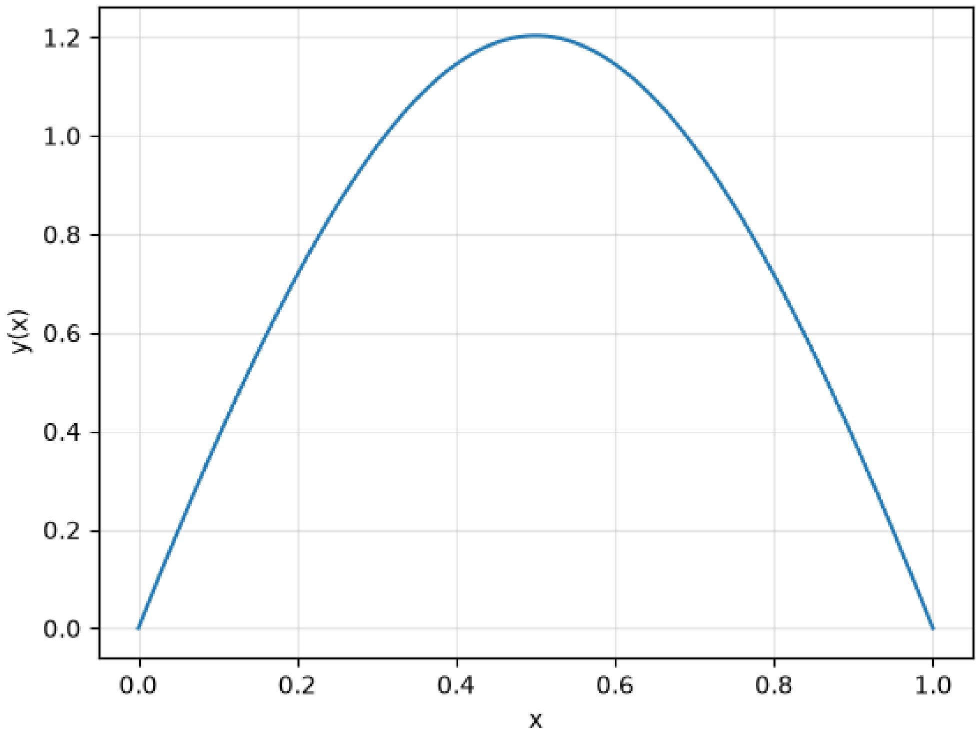
The diagnostics show:

- excellent ODE agreement,
- machine-precision boundary satisfaction,
- machine-level interface continuity,
- bounded trajectory.

### Conclusion

```
STATUS: VERIFIED
```


Visual verification    residual 6.38e-10   [VERIFIED]
both sides overlaid — they must coincide
LHS  y''
RHS  -3.5135*exp(y)
LHS alone:  y''
RHS alone:  -3.5135*exp(y)
|LHS - RHS|    max = 7.93e-10
x
x
x
x



1.2
1.0
0.8
0.6
0.4
0.2
0.0
y(x)
0.0
0.2
0.4
0.6
0.8
1.0
x

! ‘

## Case 2: Extremely Near the Fold

$\lambda = 3.5138$

### Solver Output

```
INFO: verification by substitution - VERIFIED

Equation residual:
8.36e-10

Relative residual:
8.37e-11

Boundary residual:
3.67e-10

Interface continuity:
1.5e-11

Segments:
6
```

### Interpretation

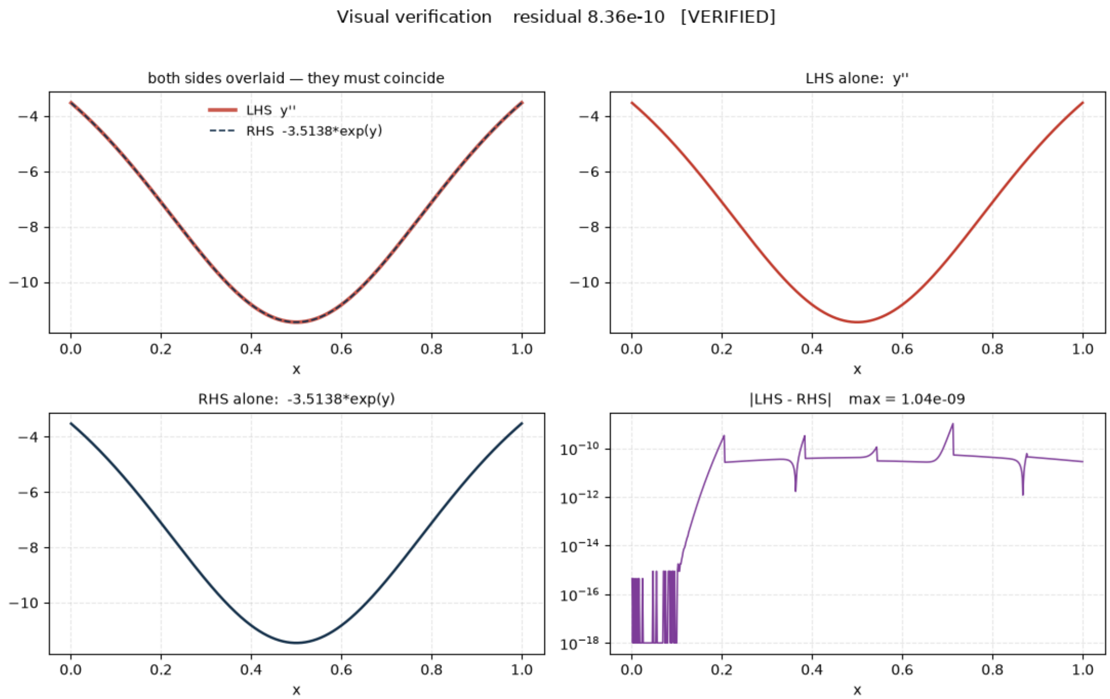
This case lies extremely close to the critical region.

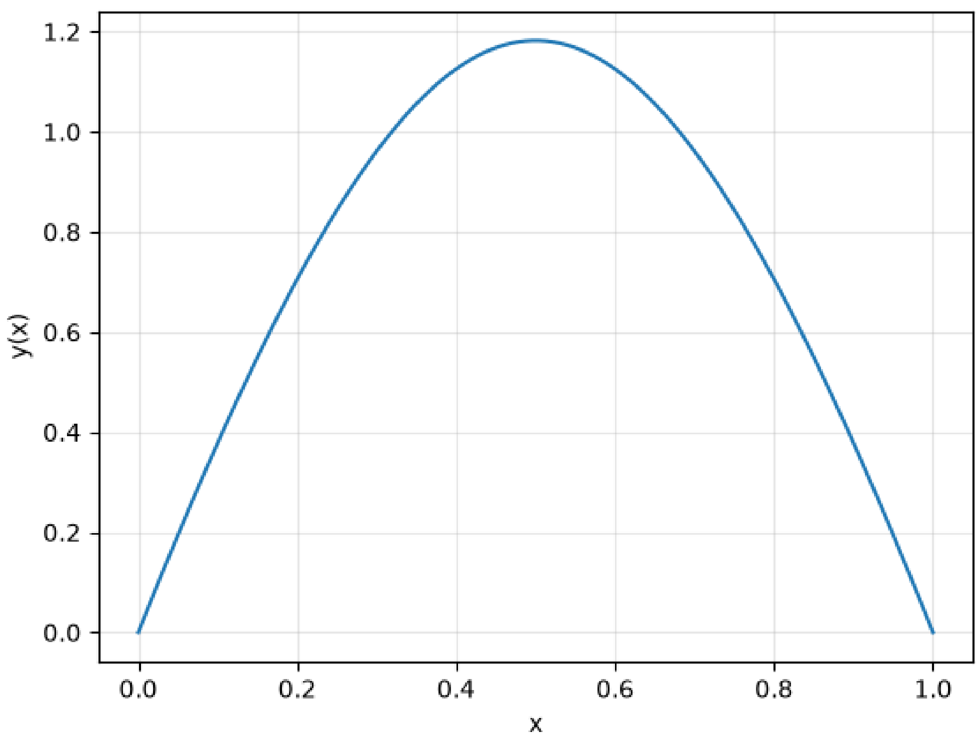
Despite the increased sensitivity:

- the solution remains verified,
- interface continuity remains excellent,
- boundary data remain satisfied.

### Conclusion

```
STATUS: VERIFIED
```


Visual verification    residual 8.36e-10   [VERIFIED]
both sides overlaid — they must coincide
LHS  y''
RHS  -3.5138*exp(y)
LHS alone:  y''
RHS alone:  -3.5138*exp(y)
|LHS - RHS|    max = 1.04e-09
x
x
x
x



1.2
1.0
0.8
0.6
0.4
0.2
0.0
y(x)
0.0
0.2
0.4
0.6
0.8
1.0
x


!

‘

## Case 3: Reported Critical Value

$\lambda = 3.513830719$

### Default Solve

The default strategy did not return a solution.

Instead SIRPY produced:

```
CERTIFICATE OF NONEXISTENCE
```

with

```
Smallest achievable mismatch:
9.291e-05

Best slope:
gamma = 3.967314

Fold location:
b*   0.999999

Distance beyond fold:
1.06e-06
```

The best trajectory itself remained verified:

```
max relative defect:
8.3e-11
```

but could not satisfy the final boundary condition.

---

### Interpretation

The nonexistence certificate indicates:

- all scanned shooting trajectories approach the boundary from the same side;
- no sign change was detected;
- the branch appears to terminate at a fold extremely close to the endpoint.

In other words:

```
The ODE was solved successfully,
but the boundary data could not be satisfied.
```

This is fundamentally different from solver failure.

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## Critical-Point Investigation

Because the critical parameter is numerically delicate, the problem was repeated with significantly larger computational resources.

```
solve_bvp(
    order=2,
    f_str="-3.513830719*exp(y)",
    x0=0,
    x1=1,
    left_bc={0:0.0, 1:"gamma"},
    right_bc={0:0},
    time_budget=50,
    iterations=30,
    partition=120
)
```

---

### Partition Solve Output

```
Partition-method BVP

120 subintervals
300 sweeps

Node matching

|y|  = 2.44e-15
|y'| = 6.64e-14
```

Fixed-point iteration itself stalled, but the interface constants were successfully reconciled through globally coupled damped Newton correction.

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### Verification

```
Equation residual:
3.55e-15

Relative residual:
3.80e-16

Boundary residual:
3.47e-17

Interface jump:
2.3e-11
```

All diagnostics returned to essentially machine precision.

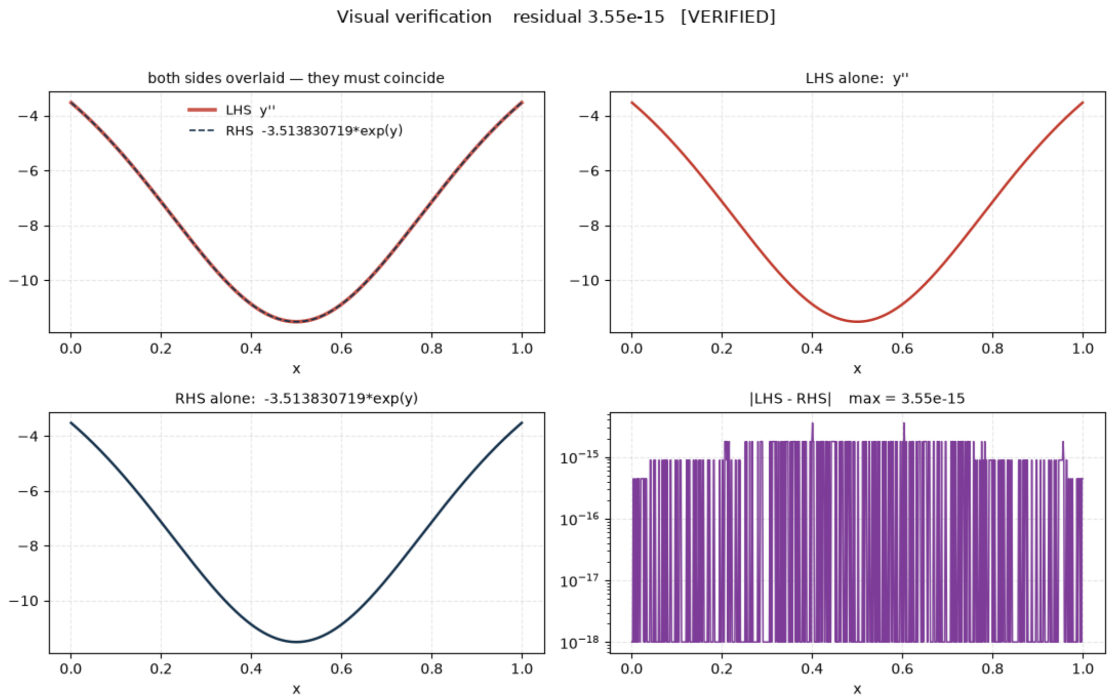
### Conclusion

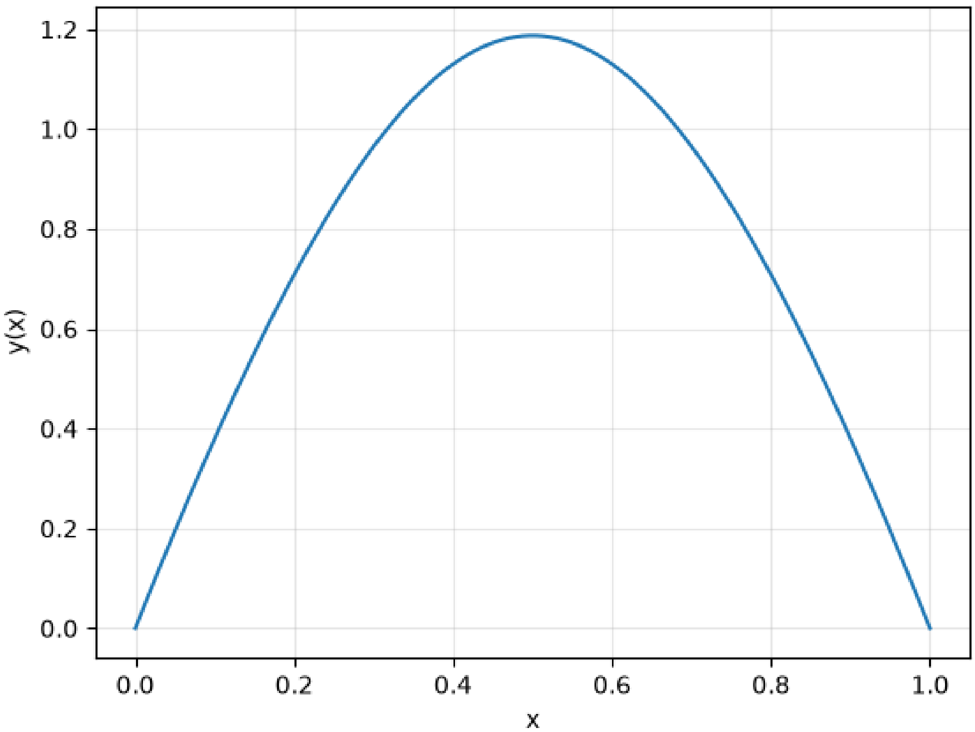
```
STATUS: VERIFIED
```

This result demonstrates that near the fold:

- the default workflow may conclude nonexistence,

- while a much more expensive globally coupled solve can still recover a verified branch.




1.2
1.0
0.8
0.6
0.4
0.2
0.0
y(x)
0.0
0.2
0.4
0.6
0.8
1.0
x

---

## Case 4: Beyond the Reported Critical Value

$\lambda = 3.52$

This case is particularly interesting.

The solver produced a piecewise construction satisfying the differential equation locally to extremely high accuracy.

### Solver Output

```
Node matching

|y|  = 9.09e-07

|y'| = 2.18e-04
```

```
WARNING

partition iteration did not fully stabilise
```

```
Interface continuity

max jump = 2.2e-04
```

---

### Local Accuracy

The local Taylor pieces remained excellent:

```
Equation residual:
3.55e-15

Relative residual:
3.81e-16
```

Boundary data were also nearly satisfied:

```
4.48e-09
```

---

## Global Consistency

The difficulty appears in the matching diagnostics:

| Quantity | Value |
|---|---:|
| Node matching in y | 9.09e-07 |
| Node matching in y' | 2.18e-04 |
| Interface continuity | 2.2e-04 |

Unlike the previous runs, the subinterval reconciliation did not fully stabilise.

---

## Why the Graph Can Look Perfect

The graph still appears visually smooth.

This is not surprising.

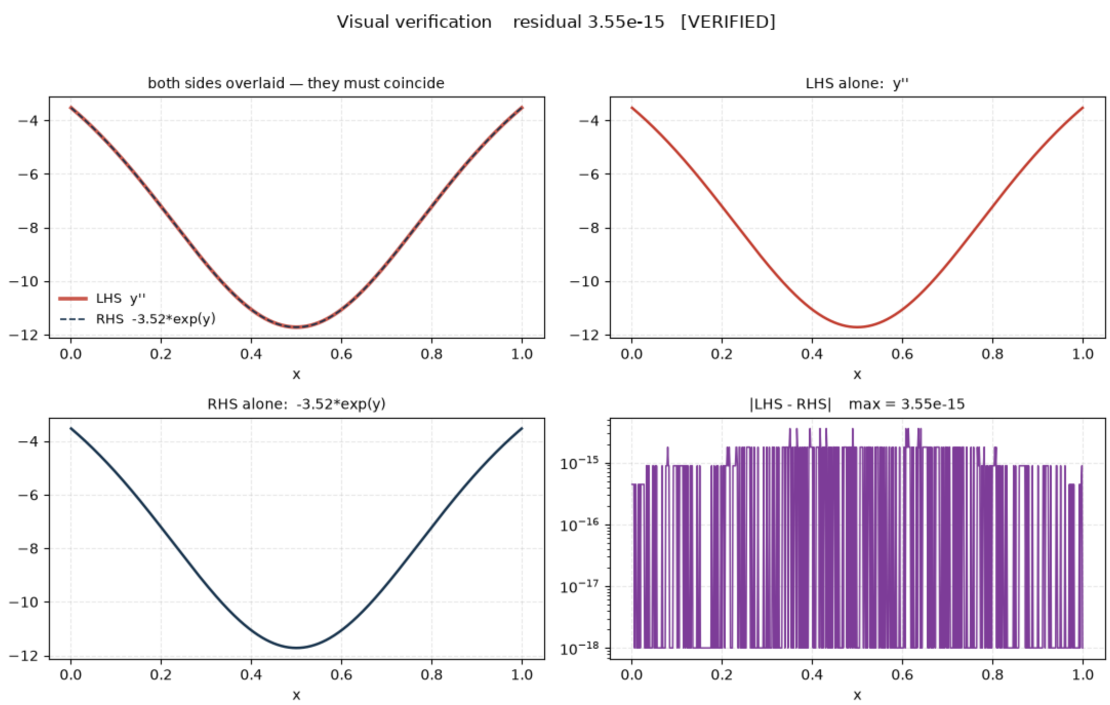
The solution itself is of order one, while the interface discontinuities are approximately

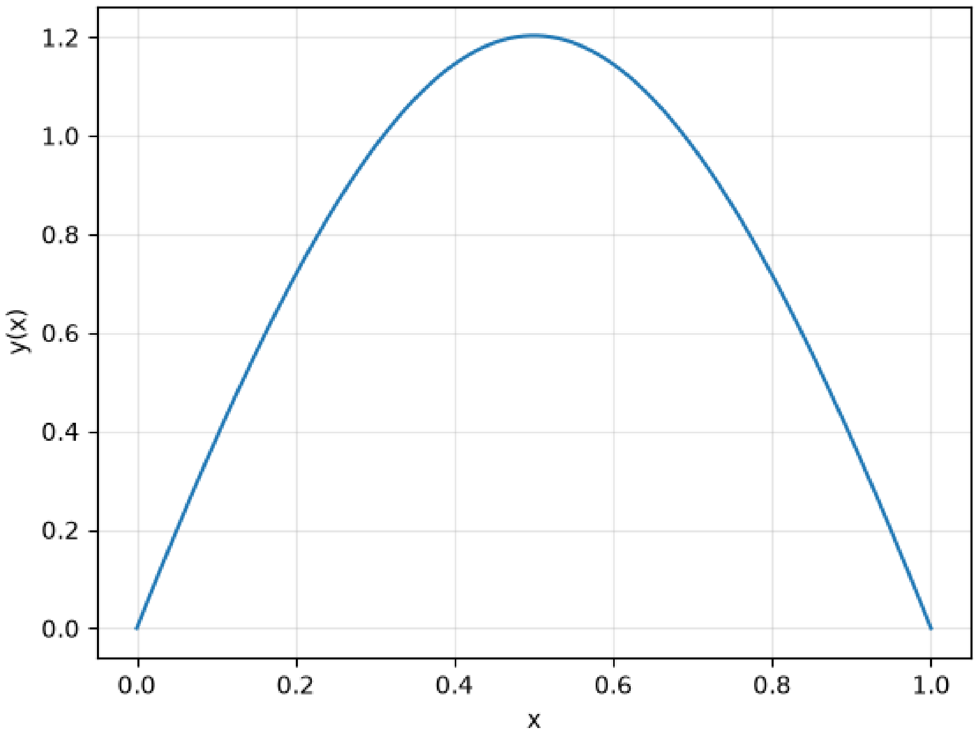
$$2 \times 10^{-4}.$$

Such defects are easily invisible on ordinary plots.

This illustrates an important lesson:

> Visual smoothness is not a verification certificate.


Visual verification    residual 3.55e-15   [VERIFIED]
both sides overlaid — they must coincide
LHS  y''
RHS  -3.52*exp(y)
LHS alone:  y''
RHS alone:  -3.52*exp(y)
|LHS - RHS|    max = 3.55e-15
x


1.2
1.0
0.8
0.6
0.4
0.2
0.0
y(x)
0.0
0.2
0.4
0.6
0.8
1.0
x

### Local Accuracy vs Global Consistency

The key observation from this benchmark is that:

- local ODE satisfaction remains outstanding,
- boundary conditions remain nearly satisfied,
- global matching quality deteriorates.

The onset of fold behavior is therefore detected first by:

```
interface continuity
```

and

```
node matching
```

rather than by the differential-equation residual itself.

---

## Benchmark Summary

| | Status | Relative Residual | Interface Jump |
|---:|---|---:|---:|
| 3.5135 | VERIFIED | 6.19e-11 | 1.6e-11 |
| 3.5138 | VERIFIED | 8.37e-11 | 1.5e-11 |
| 3.513830719 (default) | NONEXISTENCE CERTIFICATE | 8.3e-11* | N/A |
| 3.513830719 (partition) | VERIFIED | 3.80e-16 | 2.3e-11 |
| 3.52 | WARNING: global matching degraded | 3.81e-16 | 2.2e-04 |

* Best verified trajectory, not a solution.

---

## Lessons

This benchmark illustrates four distinct numerical phenomena:

1. **Successful nonlinear BVP solution**
2. **Approach to a saddle-node bifurcation**
3. **Certified nonexistence**
4. **Loss of global consistency despite excellent local accuracy**

Most importantly, it demonstrates why SIRPY separately reports:

- differential-equation residuals,
- boundary residuals,
- interface continuity,
- node matching.

Near bifurcation points these quantities can decouple.

A solution may satisfy the ODE locally to machine precision while simultaneously exhibiting deteriorating global consistency.

The Bratu problem therefore serves as a stress test not merely for solving nonlinear BVPs, but for verifying them.

### The problem library

This entry is part of the SIRPY Problem Library — an ongoing, public catalog of mathematical systems solved and verified by SIRPY.

- **GitHub** — github.com/Mitra-Institute-of-Education-MIE/sirpy-problem-library
- **Web** — mitrainstituteofeducation.org/sirpy

### Contact us

Contact: info@mitrainstituteofeducation.org · mitrainstituteofeducation.org/contact

Please only send problems you are free to share publicly.

*SIRPY is in final validation prior to its first public release. Every line quoted above is verbatim from a single run of the solver; nothing is reproduced from memory.*